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PiPo-Net: A Semi-automatic and Polygon-based Annotation Method for Pathological Images

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Abstract— **Metastatic involvement of lymph nodes is one of the most important prognostic variables for many cancers. Several deep learning based algorithms have been developed to segment metastatic regions in pathological images to help predict prognosis. However, the training of these methods requires a large amount of annotated data, and the labeling task is an extremely time-consuming process for human annotators. In order to reduce the annotation burden, we for the first time propose a semi-automatic annotation method (PiPo-Net) for the labeling of pathological images. The method is comprised of two subnetworks, a pixel-wise segmentation network (Pi-Net) and a polygon-based annotation network (Po-Net). The Pi-Net adopts an improved encoder-decoder architecture and can effectively aggregate multi-scale image features. The Po-Net is built on the Pi-Net and leverages a two-layer recurrent neural network to generate tight-bounded polygons for the metastatic regions. Corresponding to the proposed network architecture, a loss function called PiPo-loss is introduced to help optimize the whole network. The main advantage of our method is that it integrates human annotators into the prediction loop, allowing to iteratively refine the predictions according to the suggestions from human annotators. We evaluate our method on Camelyon16 database and achieve a Dice score of 91% in the initial annotation attempt. We also demonstrate the effectiveness of the human-network collaborative annotation, which achieves promising labeling results, verifying the advantages of our proposed method.**

I. INTRODUCTION

Lymph node metastasis occurs in many types of cancers, and is regarded as one of the most important prognostic variables [1]. The more the cancer cells metastasize to lymph nodes, the poorer prognosis the patient will have [2]. Hence, the detection of metastatic regions in pathological images becomes crucial in helping clinicians plan treatment. In recent years, deep learning based detection and segmentation algorithms exhibit significant advantages in locating the metastatic regions, but the training of these algorithms requires a large amount of annotated data. Since the pathological images are usually at a magnitude of gigapixels, the annotation process is quite tedious and time-consuming. In order to reduce the annotation cost while maintaining high-quality annotation results, we propose in this work a semi-automatic annotation method for labeling pathological images.

There have been many attempts on reducing the annotation burden of large-scale datasets. A commonly-used method is to draw a bounding box around the region of interest as prior information [3]–[7], and the deep neural network can then automatically generate a semantic segmentation mask inside the box. For instance, Dai *et al.* [3] propose a deep learning model that iterates between region (bounding box) proposal generation and CNN training, through which the segmentation masks are gradually recovered. Yang *et al.* [4] propose a novel bounding box based annotation model for medical image segmentation based on graph search, which is applied to localize the optimal object boundary. Khoreva *et al.* [5] employ GrabCut-like approaches to derive annotation results from the given bounding boxes. However, in order to get a tight and accurate bounding box, human annotators have to tune the box several times by carefully dragging the vertices of the box. To further reduce labeling time, point-based annotation methods [8]–[11] are proposed, in which only one or several points/pixels are needed to serve as prior knowledge for mask generation. For instance, the extreme points of an object (left-most, right-most, top, bottom pixels) are leveraged as inputs to generate the segmentation mask [8]. A time-efficient deep supervision model is proposed in [9], which combines the point supervision information and a new objectness potential in network training, to generate semantic image segmentation masks. A new auxiliary network (PseudoEdgeNet) is proposed [11], which is based on Edge Net and Attention Module to recognize nuclei in histopathology images with only point supervision. Another more user-friendly annotation method is based on scribbles [12]–[16], which only requires the annotator to drag the cursor and scribble on the interior of the objects. Specifically, the information from scribbles is propagated to unmarked pixels based on a novel graphical model.

However, there exist two serious limitations in these studies. Firstly, these methods only generate pixel-wise semantic masks. Since each pixel is predicted independently, the generated masks are often a set of disjointed regions, whereas a metastatic region is typically continuous and without holes. These pixel-wise prediction methods neglect the constraint of spatial coherency of the metastatic regions and thus easily induce noisy predictions. Secondly, if the generated annotations are scattered, it is difficult for human annotators to interactively refine the outputs. To tackle these problems, we propose to leverage polygons to semi-automatically annotate the metastatic regions in this work, and the annotation process of metastatic regions is treated as a polygon prediction task. A polygon is typically comprised of a series of connected vertices and has much lower complexity than pixel-wise masks. Hence it is an easy task for the network to predict tight-bounded polygons for the region of interests.

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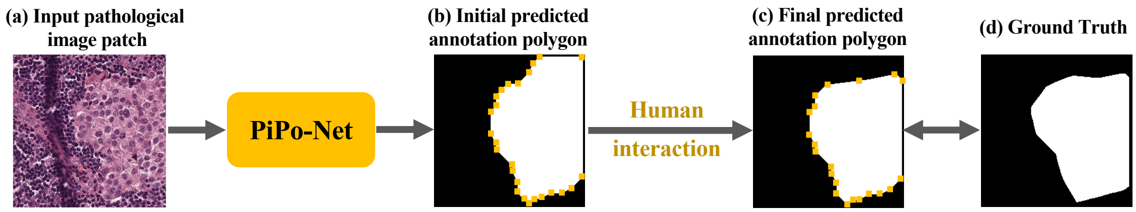

Fig. 1. Illustration of the annotation pipeline proposed in this work. (a) An input pathological image patch is first fed into the PiPo-Net to generate (b) the initial polygon annotation mask. The human annotators can interact with the initial predicted polygon (i.e., drag the misplaced polygon vertices to the right positions) in order to derive the more accurate (c) final polygon annotation. We demonstrate that the final polygon mask is much similar with (d) the ground truth, which indicates the effectiveness of the proposed method.

The risk of generating noisy predictions is also significantly reduced. Moreover, it is convenient to adjust the polygon vertices (i.e., dragging the incorrect vertices to the right positions) if the generated polygon vertices are inaccurate or polygon edges self-intersect. Taking human corrections as inputs, the network can then perform a new prediction. Such an iterative process continues until a satisfying annotation result is achieved. These good properties of polygon-based interactive annotation method can effectively address the above mentioned problems.

Our work is inspired by Polygon-RNN [17] that targets object annotations for cityscape images. The major difference with [17] lies in three aspects. Firstly, a new architecture comprised of a pixel-wise segmentation subnetwork (Pi-Net) and a polygon-based annotation subnetwork (Po-Net) is proposed, in which the intermediate outputs of Pi-Net serves as auxiliary information to assist Po-Net in polygon generation. Secondly, instead of using tight boxes bounding the region of interests as model inputs [17], which requires human annotators to first detect the objects of different sizes and then draw the boxes accordingly, here we directly crop the whole pathological images into small patches with the same resolution, and use these patches as model inputs. This operation can significantly reduce the burden of human annotators, i.e., they do not need to draw the boxes exactly bounding the objects. Thirdly, a novel loss function called PiPo-loss is introduced to further improve the annotation accuracy. Polygon-RNN takes ground-truth boundary as supervision and focuses on predicting the polygon vertices, thus neglect the importance of area constraints, i.e., the IoU between ground-truth mask and the polygon. To solve this problem, we in this work propose a loss function that additionally integrates the Dice loss to help improve the annotation quality.

Our contributions in this work are summarized as follows:

- This is the first work that proposes a semi-automatic and polygon-based annotation method for labeling pathological images. The human annotators are allowed to interact with the annotation network, by the help of which more accurate annotation results can be obtained.
- A novel architecture, i.e., PiPo-Net, is presented to help improve the annotation accuracy, in which two collaborative subnetworks are developed to conduct pixel-wise segmentation task and polygon-based annotation task, respectively. Accordingly, a new loss function is also introduced to further improve the annotation accuracy.
- The experimental results show that our method achieves a Dice score of 91% in the initial annotation attempt. We also demonstrate the effectiveness of the human-network collaborative annotation, and the promising annotation results are achieved by the guidance of human annotators.

The remainder of this paper is organized as follows. Section II explains the details of our proposed method, which has two collaborative subnetworks. Experimental results that validate the proposed model are presented and discussed in Section III. Finally, the paper is concluded in Section IV.

II. METHOD

A. Network Architecture

As illustrated in Fig. 2, the proposed network is comprised of two components, i.e., Pi-Net and Po-Net. The Pi-Net aims to produce pixel-wise segmentation masks for metastatic regions. The Po-Net is a two-layer convolutional LSTM (conv-LSTM) network and used to produce polygon annotations. Herein, the outputs of Pi-Net serves as auxiliary information to assist the Po-Net to generate more accurate polygon annotations.

Pi-Net This subnetwork adopts an encoder-decoder architecture, taking pathological image patches as inputs, and outputs pixel-wise segmentation masks. The encoder aims to extract low-level features (e.g., textures, edges) in shallow layers and high-level features (e.g., shapes) in deeper layers, and the decoder inversely map back the extracted features to predict the metastatic regions. Similar to UNet [18], skip concatenations are leveraged to connect features between each encoder layer and the corresponding decoder layer in a parallel manner (yellow dash lines in Fig. 2). Leveraging the skip connections, low-level spatial information from the encoder can be combined with high-level features from the decoder, which enables localizing the metastatic regions effectively. For both encoder and decoder blocks, a convolution operation named *Conv2d-BN-SKM-TransConv2d* [20], [21] is introduced between adjacent blocks to double the dimension of the feature maps and halve the number of feature channels. Herein, the embedded selective kernel modules (SKMs) [22] can dynamically learn informative feature representations with different sizes of kernels, i.e., 3×3, 5×5, and 7×7,

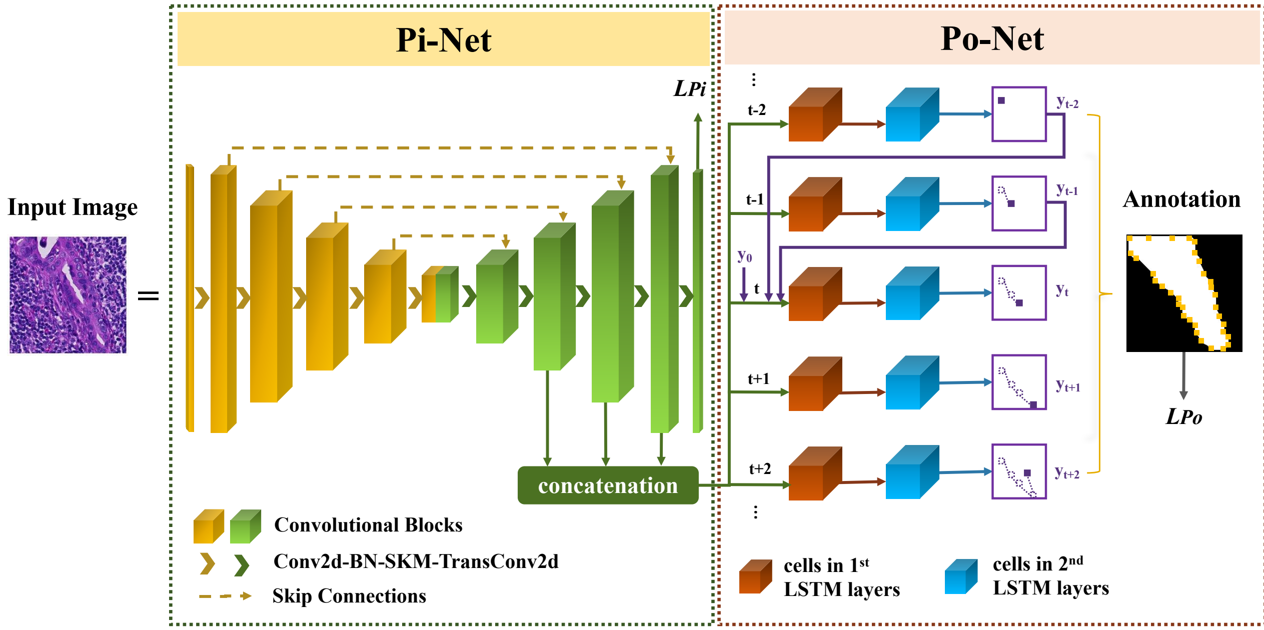

Fig. 2. Illustration of our proposed PiPo-Net. It is comprised of two subnetworks, i.e., Pi-Net and Po-Net. Pi-Net built on improved UNet [18] aims to produce pixel-wise segmentation masks for metastatic regions. Po-Net is a two-layer convolutional LSTM network [19] and used to produce polygon annotations. The input of PiPo-Net is pathological image patch, and the output of PiPo-Net is the initial polygon annotation mask (see Fig. 1b).

which significantly enrich the diversity of receptive fields and enable the network to capture more discriminative features from the input images.

The illustration of SKM can be seen in Fig. 3. Firstly, an input feature map X is convolved by three kernels (i.e., 3×3, 5×5, and 7×7), and the corresponding feature maps, i.e., X_3, X_5, X_7, are generated. Secondly, an element-wise summation is performed to combine three maps, and $\widetilde{X} = \sum X_k (k \in \{3,5,7\})$ is derived. Next, f, the adaptive feature, is obtained following the squeeze operation (*sq*) and the FC layer ($\mathcal{F}_{fc}$):

$$f = \mathcal{F}_{fc}\left(\frac{1}{H\times W}\sum_{i=1}^{H}\sum_{j=1}^{W}\widetilde{X}(i,j)\right). \tag{1}$$

Then, m_k, the weight vector corresponding to X_k, is derived as follows:

$$m_k = \frac{e^{W_k f}}{\sum e^{W_k f}} \quad (k \in \{3,5,7\}), \tag{2}$$

where W_k are learnable parameters. $\widehat{X_k}$ is generated by multiplying m_k with X_k. Finally, the aggregated feature map $\widehat{X}$ is derived by combining the adaptive features of three paths.

Po-Net The purpose of Po-Net is to generate polygon annotations, i.e., the polygon vertices, for the metastatic regions. RNN [23] is capable of processing sequential data, and modeling the long-term dependencies of input sequences. In this work, a special RNN structure, i.e., a conv-LSTM, is applied in polygon annotation task [17], [19]. The key equations of the conv-LSTM are shown in Eq. 3.

$$\begin{aligned}
i_t &= \sigma(W_{xi} * Xt + W_{hi} * h_{t-1} + W_{ci} \circ c_{t-1} + b_i)\\
f_t &= \sigma(W_{xf} * Xt + W_{hf} * h_{t-1} + W_{cf} \circ c_{t-1} + b_f)\\
o_t &= \sigma(W_{xo} * Xt + W_{ho} * h_{t-1} + W_{co} \circ c_t + b_o)\\
c_t &= f_t \circ c_{t-1} + i_t \circ tanh(W_{xc} * Xt + W_{hc} * h_{t-1} + b_c)\\
h_t &= o_t \circ tanh(c_t)
\end{aligned} \tag{3}$$

where $*$ means the convolution operation; $\circ$ denotes the Hadamard product; σ represents the sigmoid function; i_t, f_t, o_t means the input, forget, and output gate at time step t; c_t and h_t denote the cell output and hidden state at time step t; X_t means the model input; W_h denotes the hidden-to-state convolution kernel; and W_x denotes the input-to-state convolution kernel.

According to Eq. 3, we can see that the conv-LSTM replaces the internal matrix multiplications of traditional LSTM [24] with convolutional operations, which can significantly reduces the scale of training parameters and improves running efficiency. By leveraging conv-LSTM, not only the long-term time dependencies of inputs can be modeled, but also the spatial information in input-to-state and state-to-state transitions is encoded. In this polygon annotation generation task, the predicted spatial location of each polygon vertex is treated as the output of each conv-LSTM step, and the history information is also considered in the prediction to explicitly model the spatial dependencies (see Fig. 2).

Po-Net stacks two conv-LSTMs to generate hierarchical feature representations of the input. The output of each LSTM step is with a size of $H\times W$ (a 28×28 grid) and adopts a one-hot encoding schema, i.e., each grid cell is

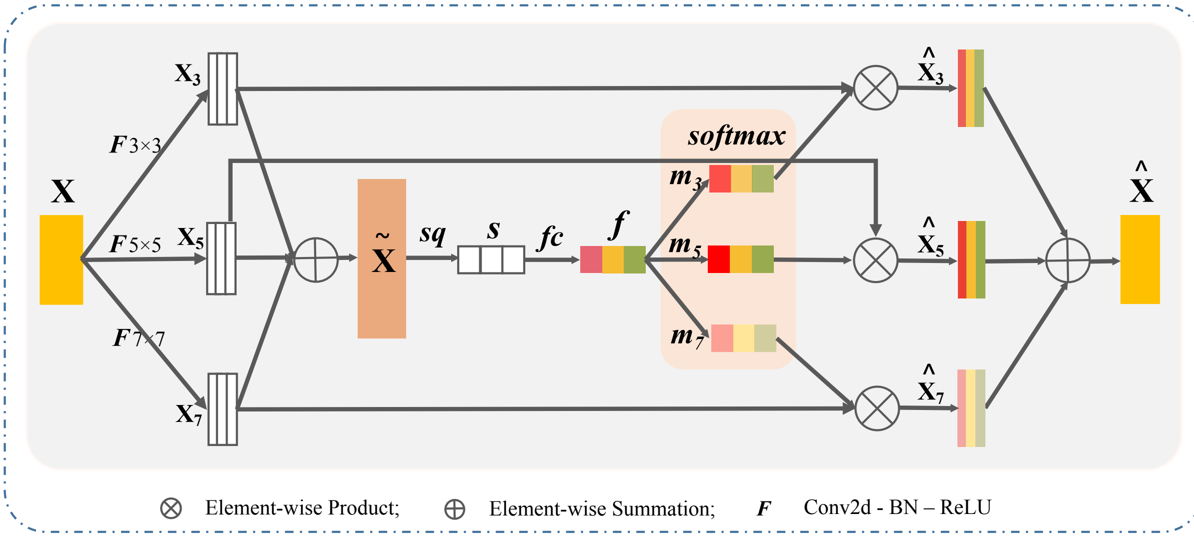

Fig. 3. Illustration of selective kernel module (SKM). X is the input feature map, which is convolved by three convolutional kernels (size is 3, 5, and 7, respectively). X_3, X_5, X_7 are the corresponding generated feature maps. $\widetilde{X}$ is the summation of these three feature maps. After squeeze operation (i.e., global average pooling) and the FC layer ($\mathcal{F}_{fc}$), the adaptive 1D feature f is obtained (see Eq. 1). m_k (k = 3, 5, 7) represents the weight vector corresponding to X_k, and $\widehat{X_k}$ is generated by multiplying m_k with X_k. Finally, $\widehat{X}$ is obtained by combining $\widehat{X_k}$ (k = 3, 5, and 7).

encoded by a 784-D vector. As shown in Fig. 2, the inputs of Po-Net at step t consist of three parts: the concatenated features from Pi-Net, the predicted vertices y_{t-1} and y_{t-2}, and an initial vertex y_0. The generation of y_0 is as follows. We adopt Pi-Net again but with two more layers added, each of which is with a resolution of 28×28. The first layer is used to predict the boundaries of the metastatic regions, and the second layer takes as input the outputs of the first layer and the concatenated feature representations derived from the decoder layers of Pi-Net, and outputs the polygon vertices. Here, both boundary prediction and vertex generation tasks are treated as a binary classification problem in each cell of the output grid. The determination of the first polygon vertex helps the conv-LSTM decide when is the end of the polygon generation procedure. During the inference procedure, each conv-LSTM step will generate a polygon vertex. By sequentially connecting these vertices, an initial polygon annotation is obtained.

For those initial generated polygons with inaccurate annotations, intervention from the human is required to help derive the final annotation mask. Specifically, for the misplaced polygon vertex, we directly change its coordinates to the correct position, and the new coordinate is fed into the trained Po-Net to replace the wrong vertex coordinate. Since output at step t depends on that at step *t-1* and *t-2*, with time step increases, a new corrected polygon is gradually formed. The algorithm of this polygon fixation procedure is shown in Alg. 1.

The reasons why the network can integrate the corrections from human annotators are as follows: 1) During the training phrase, the history information (y_{t-1}, y_{t-2} in Fig. 2) adopts ground-truth labels rather than the previous predictions, which makes the LSTM outputs sensitive to the correct annotations; 2) During the inference phrase, the previous

Algorithm 1 Polygon Fixation

Given an initial annotation $\mathcal{A}$ with misplaced vertices.
for each misplaced vertices $\mathbf{n} \in \mathcal{A}$ **do**
  $x, y \leftarrow$ GETCOORDINATE($\mathbf{n}$)
  $x^*, y^* \leftarrow$ CORRECTCOORDINATE(x, y)
  $\mathcal{A} \leftarrow$ UPDATEVERTEX($x^*, y^*, \mathbf{n}$)
  for each step t in LSTM **do**
    $x_{t+1}, y_{t+1}, \mathbf{n}_{t+1} \leftarrow$ Po-Net.PREDICT($x_t, y_t, \mathbf{n}_t$)
    $\mathcal{A} \leftarrow$ UPDATEVERTEX($x_{t+1}, y_{t+1}, \mathbf{n}_{t+1}$)
  end for
  if annotation quality of $\mathcal{A}$ is achieved **then**
    return the current annotation $\mathcal{A}$.
  end if
end for

predictions are taken as history information, thus leading to some inaccurate annotations (i.e., polygon vertices). Once these vertices are corrected to the right position and fed to the network, LSTM outputs that take these vertices as inputs will also be improved, which thus constitute a positive loop.

B. PiPo-loss Function

In this section, we detail the proposed PiPo-loss, which is comprised of L_{Pi} (loss of Pi-Net), and L_{Po} (loss of Po-Net).

Loss of Pi-Net L_{Pi} aims to measure the differences between the predicted metastatic regions and the ground-truth pixel-wise segmentation masks. L_{Pi} consists of a *binary cross-entropy loss* and a *dice loss* [25], which is as follows:

$$L_{Pi} = -\sum_i q_i log(p_i) + (1 - \frac{2\sum_i p_i q_i + \varepsilon}{\sum_i p_i + \sum_i q_i + \varepsilon}), \quad (4)$$

where p_i indicates the probability of pixel i being categorized as metastatic regions, $q_i \in \{0, 1\}$ is the corresponding ground-truth mask, and ε is a small positive number used for increasing numerical stability. The cross-entropy loss function penalizes pixel classification errors while the dice loss function measures the overlap between the predicted regions and the ground-truth area, which can handle the foreground-background imbalance problem.

One of the limitations of [17] is that the training of polygon-RNN is simply regarded as a vertex prediction process, while the importance of area constraints is neglected, i.e., the IoU between ground-truth mask and the polygon. Herein, the L_{Pi} is designed to solve this problem. L_{Pi} contains a Dice loss, and can help improve the IoU and thus the annotation quality.

Loss of Po-Net L_{Po} aims to measure the difference between the predicted polygons and the ground-truth polygons. As mentioned in Section II, each predicted polygon vertex is an one-hot encoding at each RNN step, thus the vertex prediction process can be regarded as a classification task. Here, the *binary cross-entropy loss* is leveraged to optimize L_{Po}, and the equation is similar to [17]:

$$L_{Po} = -\sum_{t} n^t log(m^t), \tag{5}$$

where m^t indicates the position of the predicted polygon vertex at step *t*, and n^t is the corresponding ground-truth coordinate of this vertex.

III. EXPERIMENTAL RESULTS

A. Dataset and Evaluation Metrics

In this work, the CAMELYON16 dataset[1] is adopted to evaluate the proposed annotation method. This dataset contains 158 whole-slide images (WSIs) with breast cancer metastasis in lymph nodes, which are divided into 110 training images, 24 validation images, and 24 testing images. All the images are cropped into 224×224 small patches with a magnitude of 10×. To reduce data imbalance problem, if the number of patches in one WSI is large, we randomly sample 1,000 from it. Patches with only one metastatic region are reserved.

The evaluation metrics we adopt are *Sensitivity, Precision, Dice Score*, and *Intersection-over-Union (IoU) Score*, which are calculated as follows:

$$Sensitivity = \frac{TP}{TP + FN} \tag{6}$$

$$Precision = \frac{TP}{FP + TP} \tag{7}$$

$$Dice = \frac{2 \times Precision \times Sensitivity}{Precision + Sensitivity} \tag{8}$$

$$IoU = \frac{TP}{TP + FP + FN} \tag{9}$$

where True Positive (TP) indicates the number of pixels that are correctly classified as metastatic regions; False Positive (FP) means the number of pixels that are wrongly classified as metastatic regions; and False Negative (FN) denotes the number of pixels that are wrongly classified as the background.

In all experiments, a mini-batch of size 8 is used and the Adam optimizer [26] is adopted for model training. The initial learning rate is set to 1e-4 and decreased by a factor of 10 after 10 epochs, with a total of 20 training epochs. All models are implemented based on the PyTorch framework [27] with a 24 GB memory NVIDIA Tesla P40 GPU card.

B. Comparative Experiments

In this section, some ablation studies are conducted to evaluate the performance of our method. The experimental results are summarized in Table I.

Models in row 1 and 2 are trained with L_{Po}, and the difference between them lies in the feature extraction module.

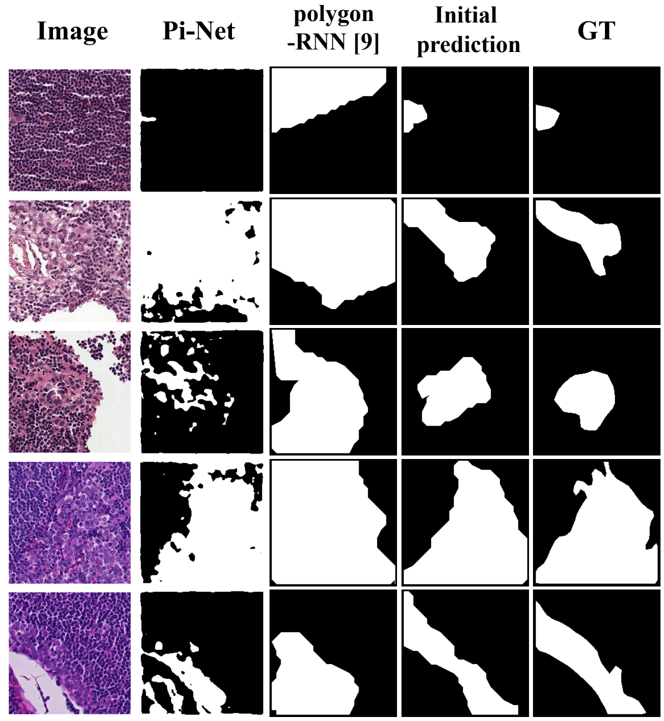

Fig. 4. Comparison of predictions of different methods and the initial polygon annotation masks generated by PiPo-Net on Camelyon16 dataset. The prediction outputs of the pixel-wise method, i.e., Pi-Net, are a set of disjointed regions. The constraint of spatial coherency of the metastatic regions is neglected. By comparing generated outputs of polygon-RNN [17] and our proposed PiPo-Net, we demonstrate the effectiveness of our method.

TABLE I
TESTING RESULTS OF ABLATION STUDY ON CAMELYON16 DATASET.

Model	Sen	Pre	$Dice$	IoU
polygon-RNN [17]	0.84	0.90	0.83	0.78
PiPo-Net w/ L_{Po}	0.93	0.90	0.90	0.84
PiPo-Net w/ $L_{Pi} + L_{Po}$ (Ours)	0.94	**0.91**	**0.91**	**0.85**
Pi-Net	**0.95**	0.89	0.89	0.84

[1]https://camelyon16.grand-challenge.org/

Polygon-RNN (row 1) [17] adopts VGG-16 framework as the feature extractor, while model in the 2^{nd} row adopts a SKM-embedded UNet (i.e., Pi-Net) to capture feature representations. The significant improvement of Dice and IoU score indicates the superior performance of Pi-Net than [17]. Specifically, SKMs in Pi-Net enrich the diversity of receptive fields and help the network to capture more discriminative features from the input images, and the encoder-decoder architecture with skip connections enables the fusion of both contextual (high-level) and local (low-level) information.

L_{Po} aims to measure the difference between the predicted polygon vertices and the ground-truth vertices, thus the optimization of L_{Po} only considers the outline/boundary of the polygons, and neglects the information provided by interior areas. Therefore, L_{Pi}, a pixel-wise loss function for metastatic region segmentation, is introduced to provide more supervision information. We hypothesize that the combination of area (L_{Pi}) and boundary (L_{Po}) supervision can enable more accurate annotations for the following reason. The pixel-wise segmentation task and the polygon-based annotation task collaborate with each other, and the joint optimization enables the feature extraction module to learn more general features that are shared by both tasks. If the entire network is only trained with L_{Po}, it can be easily trapped in a single feature space, which may increase the risk of overfitting. The experimental results comparing the 3^{rd} against the 2^{nd} row show the effectiveness of this collaborative loss function. We also compare the proposed method with a pixel-wise segmentation method, i.e., Pi-Net (row 4), which achieves a Dice score of 0.89, and an IoU score of 0.84. Compared with this pure pixel-wise segmentation method, it can be seen that our model improves the Dice and IoU score by 2.25% and 1.19%, which indicates that our annotation method can achieve high-quality initial annotation results. We demonstrate the comparison results of different methods in Fig. 4. Moreover, in many cases, annotations generated by our methods are more accurate than those labeled by pathologists. Some examples are displayed in Fig. 5. As we can see, these samples are very challenging, and the proposed method can help save much annotation cost.

C. Interactive Annotation

Our method allows human annotators to participate in the prediction loop, in which the network predictions are iteratively refined according to suggestions from human annotators. Particularly, at each RNN step, the human annotators can correct a misplaced vertex, and the corrected vertex is then fed to the model to replace the previous incorrect prediction, which can effectively help the model to get back to the right track. Several cases showing the interactive annotation are demonstrated in Fig. 6. We correct a vertex if it deviates from the ground-truth vertex by a certain distance. Columns 3-5 show the results after one, two, and three iterations, including the correction from the annotators and re-prediction from the network. It can be seen that as the number of annotation interactions increases, the promising annotation results are obtained.

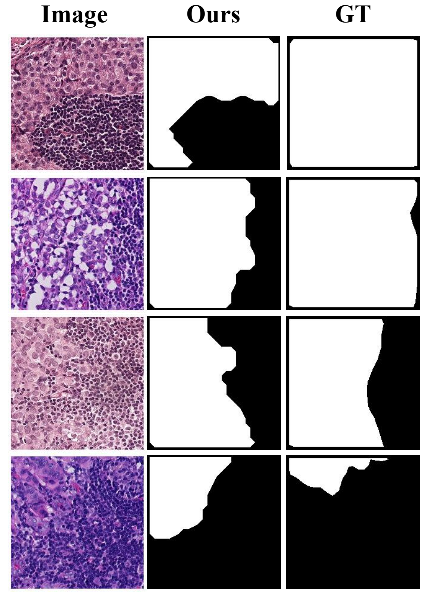

Fig. 5. Examples showing that our predictions are more accurate than pathologists' annotation, i.e., Ground Truth (GT).

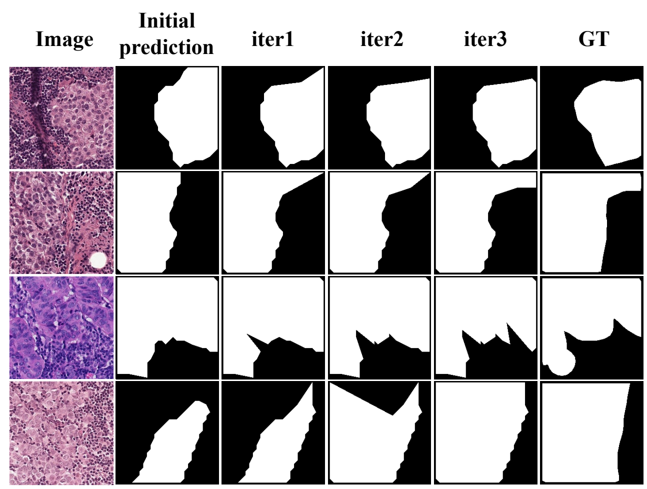

Fig. 6. Demonstration of interactive annotation of several cases with poor initial prediction performance. As shown, with iteration step increases, the predicted polygon becomes more accurate and much close to GT.

IV. CONCLUSION

In this study, we for the first time to propose a semi-automatic annotation approach to generate tight-bounded polygons for pathological images, which effectively aggregates features extracted from two collaborative subnetworks, i.e., Pi-Net and Po-Net. A novel loss design that considers the dependency between semantic segmentation and polygon annotation task is also introduced to further boost the annotation accuracy. Our method allows the human annotators to intervene the annotation results and iteratively refine the wrong predictions. Experiments demonstrate the effectiveness of our proposed annotation method, and the promising performance of human-network collaborative annotation.

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